

Which DTH TV for You?

The article 'Which DTH TV for You?' published in February issue covered general information about the DTH TV technology and services. I feel the actual service of a DTH agency could only be experienced by a user after the system has been installed, incurring the initial expenditures and subscription costs. Even if the consumer is not satisfied with the services, he has no option but to bear with the terms and conditions of the service provider.

I was expecting EFY to have closely studied various aspects of the DTH service providers and tested the relevant standards and reports, so that the consumers could have a clear idea of the available DTH services. Almost all the DTH system agencies in India have coverage over the entire country and offer similar packages at all places.

Also, to have a good control over DTH service providers regarding the subscription/recharging system, packages offered, validity periods, etc, a regulating agency needs to be established by the government so that their monopoly is put under check.

Suman Prakash
Through e-mail

Plasma Arc Gasification

In the 'Plasma Arc Gasification for Waste Management, Part 2' article published in February issue, it is mentioned on page 126 that Westinghouse is a strategic partner of The Solena Group. I would like to bring it to your attention that this information is incorrect.

Wanda Posehn
Director, Corporate Communications
Westinghouse Plasma Corporation,
Alter NRG

The author Gp Capt. K.C. Bhasin replies:

My information was based on:

1. A report prepared by Jessica L. Wargo, Massachusetts Institute of Technology, in June-August 2002 for U.S. Environmental Protection Agency, Solid Waste and Emergency Response Technology Innovation Office, Washington D.C.

2. A '.ppt' document available on

the Internet in which Solena Group referred to Westinghouse Plasma Corp. as their 'Plasma Equipment, Design and Integration Partner'. I understand that WPC, at some point, severed that relationship and Solena is now working on their own plasma torch designs.

Now that NRG (The Nuclear Research and Consultancy Group) has confirmed that WPC does not have any strategic partnership with Solena Group, I honour their word.

Software in EFY-CD

The EFY-CD contains lots of good software. Could you also include some advanced design and simulation software, like the one provided by National Instruments?

Zarinayasma
Chennai

Mobile Bug

Regarding the 'Mobile Bug' circuit published in Jan. 2008 issue of EFY, I want to know:

1. What are capacitors C2 and C3 used for?
2. Up to what level does this circuit work?

Please also detail the PCB layout and other circuit-related information.

Manish
Through e-mail

The author D. Mohan Kumar replies:

This circuit is intended to detect unauthorised use of mobile phones. It also helps to detect unauthorised video and audio recordings. It detects the signal from mobile phones even if it is kept in the silent mode. It also detects SMS.

A capacitor has two electrodes separated by a dielectric, such as paper or mica. The non-polarised disk capacitor is used to pass AC and not DC. It can store energy and pass AC signals during discharge. A 0.22 μ F capacitor is selected because it is low value and has large surface area to accept energy from the mobile radiation.

To detect the signal, the capacitor is arranged as a mini loop aerial. To make the loop aerial, leads of the capacitor should be long enough. A length of 18 mm is selected by trial and error. Lead

spacing of 8 mm is also fixed to make the loop aerial.

In short, with this arrangement, the capacitor works like an air-core coil with the ability to oscillate and discharge current.

One lead of the capacitor gets DC from the positive rail and the other lead goes to the negative input of IC1. So the capacitor gets energy for storage. This energy is applied to the inputs of IC1 so that the inputs of the IC are almost balanced with 1.4 volts. In this state, the output is zero. But at any time, the IC can give a high output if a small current is induced to its inputs.

When the mobile phone radiates high-energy pulsations, the capacitor oscillates and releases energy to the inputs of the IC. This oscillation is indicated by the flashing of the LED.

C2 is a bypass capacitor for the antenna. The antenna is not necessary if capacitor C3 protrudes from the casing of the circuit. The prototype version has only a limited range of 1.5 meters. **EFY:** Since the mentioned circuit was printed under 'Circuit Ideas' section, the PCB layout is not available.

Speedometer-cum-Odometer

I want to assemble the 'Speedometer-cum-Odometer' project published in EFY's Nov. 2008 issue. If the wheel rotates in anti-clockwise direction, will the display show the value for anti-clockwise direction?

Krishna Raja Perumal
Through e-mail

The author Arun Kumar replies:

The project uses a simple transducer (reed switch and magnet) to measure the distance and speed. It does not incorporate direction detection. It is assumed that the motorbike moves in forward direction. The anti-clockwise rotation of the wheel does not decrement the odometer reading.

No Pink Slip

Relaytronics has just changed its name to Evolute. So readers may please read its name as Evolute in 'No Pink Slip, But Hiring Under Scrutiny' article published in this issue on pages 36 and 38.